Classification of contact type based on the magnetic field data for tactile sensor

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Abstract. This paper presents a machine learning approach for real time contact geometry classification for an magnetic tactile sensor array used in a robotic hand. Tactile sensing capabilities are essential for dexterous manipulation tasks that humanoid robots need for entering the workforce. For testing the developed sensor a reproducible methodology was developed for classifying point, line and surface contacts. An array of 16 3D Hall-effect sensor is used to detect the change in the magnetic field as the magnets that are suspended in silicon are displaced due to contact on the sensor surface. To collect data the framework employs a CNC milling machine that is used to systematically press custom designed contact profiles against the sensor array. Multiple classifiers architectures are evaluated offline to demonstrate the adaptive nature of the proposed methodology. Results demonstrate that narrow neural network classifier with a single hidden layer od 10 neurons can achieve optimal performance, reaching 99.5% validation accuracy for three-class classification (point, line, surface) and 99.1% for four-class classification (including no contact), with a compact model size of 15 kB suitable for microcontroller deployment.

Keywords: tactile sensing, magnetic sensors, contact classification, machine learning

1 Introduction

Human beings rely deeply on the sense of touch when manipulating objects. This sense, that is incorporated into the skin and the nervous system[1], provides valuable information for dexterous manipulation of objects including its shape, texture and temperature. In robotic systems, to compensate for the lack of information we often use controlled environments and setup tasks so that the robots can manipulate the objects without tactile sensing. To achieve a dexterous manipulation sens of touch needs to be more than just sensing the force that is applied [2]. Recent development of AI has started to fuel a rapid development of humanoid robots and their incorporation into the workforce. Many tasks, like handling delicate or deformable objects require robots to have a sense of touch that can provide additional feedback for adaptive manipulation. Another problem are cluttered objects where it is important to detect the slippage that can

occur if the object catches upon another object. Researchers have been tackling these tasks for a while, but the recent events have put this topic into the spotlight. Some estimates [3] are that the sensory components market size will increase in the next 3 years more than 10-fold, and in the next 6 years about 20 times the current market size.

There are many application of machine learning in tactile sensors [4], and new research is expanding the range of applications [5, 6]. This work focuses on determining the contact geometry for a robotic hand tactile sensor. Specifically classifying point, line and surface contacts based on the spatial response of the magnetic sensor array. Knowing which type of contact has been established can help higher levels better understand how the object touches the robot and how to approach things like slip mitigation and compliant control. We are exploring the tactile sensor [7] array that uses hall-effect sensors and magnets that are embedded into silicone. While there are other studies on this topic [8, 9], we are trying to develop an affordable, durable and reliable solution that can be used to give tactile feedback for robotic fruit pickers.

The main idea is to use raw data from the sensors and offline training to create a classifier that can be deployed to the microcontroller providing real-time inference. To achieve this we present a reproducible data collection and classifier exploration methodology that can be used for different sensors or even different classification problems.

The remainder of the paper is organized as follows: First, a brief presentation of the sensor array board and the data collection setup, next the data collection and preprocessing. Afterwords the offline classifier training is explained and the results achieved. Lastly, we provide a conclusion from our research and discuss possible further work.

2 System setup

The magnetic tactile sensor was built around TMAG5170, 3D Hall-effect sensors by Texas Instruments [10]. These are digital sensors that communicate with a microcontroller using serial peripheral interface (SPI) communication, and are configurable with different measurement ranges and operational modes to optimize sensitivity and sample rate. Sensors are placed in a 4 by 4 grid that covers an area of 20 by 20 millimeters. Spacing between the sensors is equal on both the X and Y axes and the sensor array is placed on a custom PCB board that has additional chip select multiplexing circuitry to lower the needed connections to the microcontroller.

To simulate different types of contact 3D printed probes were created using, additive manufacturing, that are attached as tools to the CNC milling machine. The probes ware design in such a way to eliminate metallic components from the milling machine in the vicinity of the sensors. Figure 1. on the left presents the three distinct contact types, point, surface and line that are made out of polylactic acid thermoplastic polyester. The point probe has a diameter of 6.15 mm while the surface probe has a diameter of 12.3 mm. The line probe has

rectangular profile of 20 mm by 4 mm. The outer layer of the sensor was also created using additive manufacturing, but with an thermoplastic polyurethane, which provides elasticity.

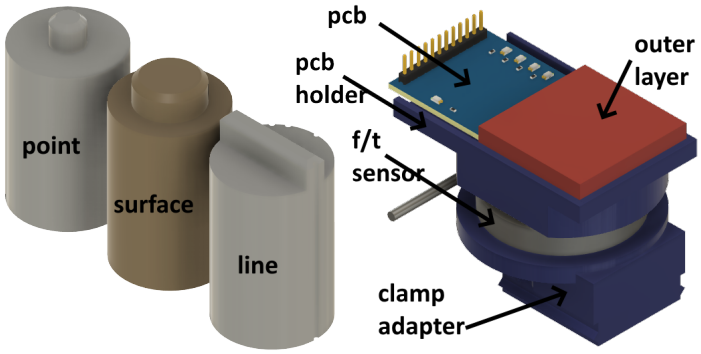


Fig. 1. Contact profiles of different probes and the custom holder with the whole assembly

The magnetic tactile sensor array was mounted on a custom holder, shown on the right in Fig.1, designed to securely hold the PCB in a fixed position and accommodate a referent force/torque sensor beneath the sensor array. There is also a clamp adapter that is used to clamp the whole setup on to the CNC milling machine. For this classification problem the force/torque sensor data is not used in any way and only data from the magnetic sensor array is used. Figure 2. shows the real setup with the line contact in place for acquisition.

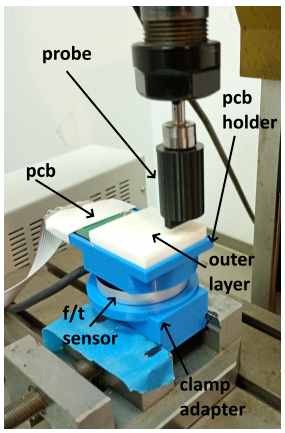


Fig. 2. Data acquisition setup

The setup is placed on a CNC milling machine and the milling machine is programmed to press the magnetic tactile sensor at different points. It uses one of the digital outputs to send signals over UART to a computer that is collecting the data. This signal represents that the milling machine is in place and the acquisition can start. The CNC then waits a certain amount of time before moving to the next point. For training data, a grid is created starting from 0,0 and moving in 1 mm increments along the X and Y axes. A total of 441 points are used for training, and with 50 samples per point, a total of 22 050 samples were collected. All data recorded during acquisition is saved in a CSV file. For validation testing a separate grid is used starting at 0.5, 0.5 and moving in 1 mm increments with a total of 400 points, meaning 20 000 samples in total. These points are used after training is completed as unseen data.

Collected data are examined to determine the validity of the data. Since the computer timer is used as one of the logged pieces of information, irregularities in the time can be a signal that something was not logged correctly. Both the magnetic sensor array and the referent force/torque sensor send out their timers that are also logged. Lastly the state of the CNC digital output is also recorded to see if the data is logged at the correct time while the CNC is pressing the sensor. Figure 3. shows some of the signals that are collected for the three different probes. Since there are 16 sensors and 3 axis per sensor only a few representative signals are shown from the training grid combined in to one plot.

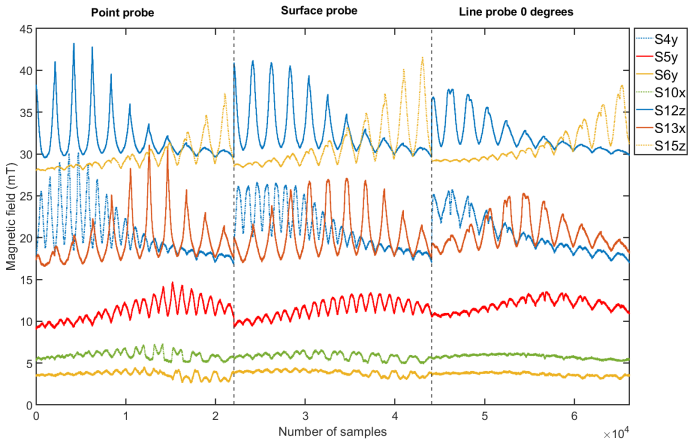


Fig. 3. Some of the signals from sensor array for different probes

Figure 4. shows some of the representative signals collected from the training grid when the line contact is at 0 degrees, and the same signals from the validation grid when the line contact is at 45 degrees.

After confirming the validity of the data, it is imported to MATLAB for preprocessing and classification learning. Preprocessing includes data extraction and data consolidation. A training part of the dataset was created by combining all magnetic data from all contact types that were collected using the training grid. The validation part of the dataset was created using all the magnetic data

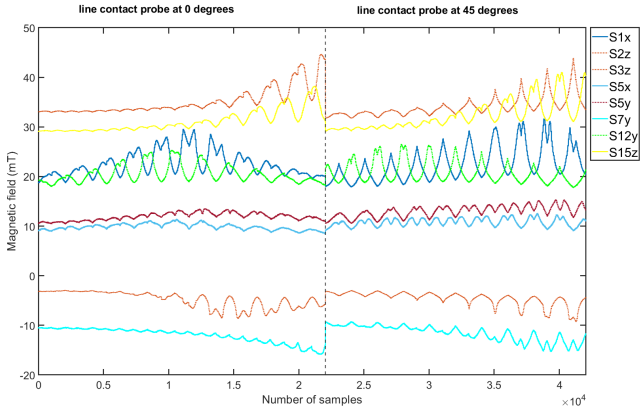


Fig. 4. Line probe signals from the sensor array

from all contact types that were collected using the validation grid. For both training and validation a class label variable is created for each sample.

3 Classification Framework

Data was imported to the Classification Learner to evaluate multiple classifier types. This toolbox enables rapid prototyping of different algorithms and automated hyperparameter tuning as well as standardized performance metrics and visual comparison. Most classifiers trained have a high accuracy rate, more than 87.5 percent accuracy, and first the classifiers that have lower accuracy than that are excluded. Next important part is the size of the model, since we want this to be deployed to a microcontroller it is important that the classifier chosen does not have a large model size. Table 1. lists different types of classifiers and the number of different classifiers trained as well as the minimal and maximal model size and the achieved accuracy for both training data and validation data.

Table 1. Comparison of Classifier Types

Classifier Type	No. Models	Model Size (Min–Max)	Best Accuracy (Train / Val)
Decision Tree	3	10 kB – 30 kB	99.9% / 96.8%
Discriminant Analysis	2	45 kB – 115 kB	100% / 99.9%
Naive Bayes	2	30 kB – 91 MB	99.4% / 98.9%
Support Vector Machine	6	40 kB – 4 MB	100% / 99.6%
Nearest Neighbor	6	24 MB	100% / 99.6%
Ensemble	5	445 kB – 394 MB	100% / 99.3%
Neural Network	5	15 kB – 49 kB	100% / 99.9%

A total of 29 classifiers in 7 different types of classifiers were tested. And as seen in Table 1. Nearest Neighbor classifiers are large in size and are not applicable for our application on a constrained environment. Out of all the types the Decision Tree Classifiers and the Neural Network Classifiers are the optimal solution for the problem and we will look at them in detail in the next chapter.

4 Results

After the initial testing two datasets were created. Dataset A has 3 classes: point contact, surface contact, and line contact. For training part of the dataset, the training grid was used and the line contact was at 0 degrees in reference to the Y axis. The validation (testing) was done using data from the validation grid where the line contact was set to 45 degrees. So not only are the different points of contact on the tactile sensor but the line contact is at a different orientation. Dataset B has 4 classes the three classes of Dataset A and added no contact samples so that the tactile sensor can distinguish between types of contact or no contact.

Table 2. Comparison of Decision Tree and Neural Network Classifiers

Variant	Classes	Configuration	Model Size	Accuracy (Train / Val)
Decision Trees				
Fine	3	100 splits	30 kB	99.9% / 94.2%
Medium	3	20 splits	15 kB	99.3% / 94.1%
Coarse	3	4 splits	10 kB	93.8% / 91.7%
Fine	4	100 splits	33 kB	99.9% / 91.8%
Medium	4	20 splits	16 kB	99.2% / 95.1%
Coarse	4	4 splits	10 kB	93.7% / 91.5%
Neural Networks				
Narrow	3	1/10	15 kB	100% / 99.6%
Medium	3	1/25	21 kB	100% / 99.9%
Wide	3	1/100	49 kB	100% / 99.8%
Bilayered	3	2/10/10	17 kB	100% / 98.8%
Trilayered	3	3/10/10/10	19 kB	100% / 99.6%
Narrow	4	1/10	15 kB	100% / 99.1%
Medium	4	1/25	21 kB	100% / 99.8%
Wide	4	1/100	50 kB	100% / 99.9%
Bilayered	4	2/10/10	17 kB	100% / 98.4%
Trilayered	4	3/10/10/10	19 kB	100% / 96.9%

For the decision tree classifiers the granulation can be defined and we have three distinctive models coarse, medium and fine depending on the maximum number of splits that a tree can have. This hyperparameter can also be optimized to achieve better results or to balance between accuracy rate and model size. Here we didn't explore optimization but this is something that can be explored in the future. Results of the testing are presented in Table 2. The results show that the decision tree classifier size depends heavily on the number of splits, which is consistent with theoretical expectations. Besides that we can see that the different orientation of the line contact lowers the accuracy rate of all classifiers.

From the results we can determine that the best solution is the medium tree classifier, which is not too large, but also not overtrain on known data so that it can detect a line contact even if it has different orientation with the best accuracy of the three models. The same datasets were used to test the neural network

classifiers. Here we have 5 classifiers, three that have only one hidden layer, one that has two fully connected layers and one with three fully connected layers. The Narrow Neural Network Classifier uses one hidden layer with 10 neurons, while the Medium has 25 neurons and the Wide Classifier has 100 neurons in one hidden layer. We can see that the increase in neurons increases the size of the classifier, which is logical and there has to be a tradeoff depending on the resources available on the microcontroller. The results are shown in Table 2. We can say that all the classifiers achieve maximal accuracy on trained data, but the validation part of the datasets with the line orientation change introduces small losses in accuracy. The smallest Neural Network Classifier offers validation accuracy of 99.6 percent for Dataset A and 99.1 percent for Dataset B, with a size of only 15 kB, providing the optimal solution.

If we look at the confusion matrix of the optimal solution, as shown in Fig. 5., it shows that the loss in accuracy comes from wrongly predicting samples from class 2 to class 3 and 1. Class 1 is the point contact, and Class 3 is the line contact. The error occurs when classifying the surface contact as one of the other two contact types. It should be noted that the normal force recorded on the force/torque sensor for all the probes was around 15N, and different forces are something that needs further exploration.

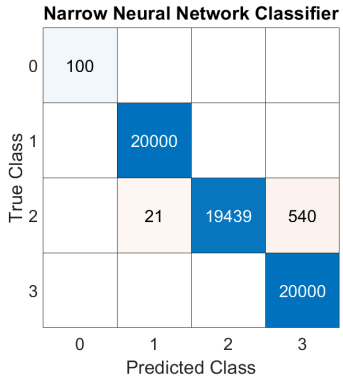


Fig. 5. Confusion Matrix for Narrow Neural Network Classifier with 4 classes

5 Conclusion and Further work

The presented approach to determining the type of contact in a magnetic tactile sensor array has proven to be feasible. Leveraging the offline training it is possible to achieve a high-accuracy classifier that can be deployed on a constrained environment to provide information without requiring additional computational power. The presented methodology can be used for other classification problems using different sensors. The optimal solution for this problem was the small neural network classifier that achieved satisfactory results even with unknown data, proving that the neural network has learned how to classify raw magnetic readings instead of just memorizing them. This enables a tactile magnetic sensor to

determine the type of contact providing information that is needed for dexterous object manipulation. Further work will involve exploring real-time classification on the microcontroller and different pressures to create an reliable tactile sensor that can determine accurately which type of contact is applied to it. Allowing robots to better understand the world around them.

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